

A Multi-Axis Robustness Analysis of Retrieval-Augmented Generation for Multi-Hop Requirements Traceability

Anonymous Author(s)

Anonymous Institution

Abstract. Published comparisons of GraphRAG and vector RAG reach conflicting verdicts, and each typically rests on a single corpus, embedder, and judge. We hold five RAG pipelines fixed and vary the embedder, the corpus (DO-178C requirements, MuSiQue, and 2WikiMultihopQA), the graph traversal, and the faithfulness judge, over 13,456 generation runs and more than 30,000 judgments. We find that the point at which citation quality is measured decides the comparison. GraphRAG’s graph walk fills the context with mostly irrelevant chunks, at a precision of 0.12 to 0.23, while its synthesizer cites selectively. Scored on the retrieved context, GraphRAG therefore ranks below the vanilla baseline in all six settings we test; scored on the answer’s citations, it ranks first. The best pipeline at each hop depth is stable across embedders and seeds. Faithfulness declines with hop depth on all three corpora once the comparison is adequately powered, and the decline follows context expansion: the one pipeline that does not expand its context does not decline. LLM judges, in contrast, are unstable. On identical inputs re-judged eleven weeks later, self-agreement falls to $\kappa \leq 0.14$, and two vendors reverse which of them is stricter between corpora. We recommend that RAG architecture comparisons report the citation measurement point and be repeated across these factors.

Keywords: Retrieval-augmented generation · GraphRAG · Requirements traceability · Multi-hop QA · LLM-as-judge · Evaluation methodology

1 Introduction

Software traceability is the ability to relate each requirement to the artifacts it derives from, satisfies, and is verified by. For airborne software, DO-178C makes this relation a certification objective, and certification authorities increasingly expect trace chains to be auditable across two or more hops [7, 8, 24, 26]. Because requirements management tools record these links explicitly, a requirements corpus is also a typed link graph, and answering a traceability question over it is a retrieval problem over both text and graph.

Retrieval-augmented generation (RAG) is a natural fit for this task, but published comparisons of RAG architectures disagree. Microsoft GraphRAG [6]

exploits graph structure, yet loses to vector RAG on detailed queries in recent benchmarks [15, 31]. Adaptive-RAG [17] routes queries by predicted complexity but does not consider hop distance, and traceability systems such as LiSSA [10] and TVR [23] do not reason over typed graphs. Each of these evaluations reports which architecture wins in one setting, usually with one corpus, one embedder, and one judge. They do not explain why the verdict changes when the setting changes.

In this paper, we hold a matrix of five pipelines fixed (vanilla, agentic, agentic with a graph tool, GraphRAG, and a learned adaptive router) and vary four factors around it. These are the retrieval embedder (e5-small, OpenAI text-embedding-3-small, and bge-m3, from three vendors and at three dimensionalities), the corpus (DO-178C requirements with typed edges, and Wikipedia paragraph chains from MuSiQue and 2WikiMultihopQA), the graph traversal (a bounded two-hop walk and personalized PageRank), and the faithfulness judge (GPT-5.4, GPT-4.1, and Gemini 3.7-flash). The design comprises 13,456 generation runs and more than 30,000 faithfulness judgments. We ask four research questions. **RQ1**: Which pipeline produces the best answer-level citations at each hop depth, and is that ordering stable across embedders, corpora, and seeds? **RQ2**: Does the GraphRAG versus vector RAG comparison depend on whether citation quality is measured on the retrieved context or on the answer? **RQ3**: Does faithfulness decline with hop depth on every corpus, and which property of a pipeline determines the decline? **RQ4**: How stable are LLM faithfulness judges across time, retrieval states, and vendors?

We find that the best pipeline depends on hop depth but not on the embedder or the seed (RQ1); that GraphRAG ranks below the vanilla baseline when citations are scored on the retrieved context and first when they are scored on the answer, in all six settings we test (RQ2); that faithfulness declines with hop depth on all three corpora once adequately powered, and only for pipelines that expand their context (RQ3); and that judges drift on identical inputs and reverse which of them is stricter between corpora, while reproducing the same hop-wise ordering (RQ4).

2 Related Work

Agentic and adaptive RAG. Self-RAG [1] and CRAG [32] introduced reflective and corrective retrieval. Adaptive-RAG [17] routes queries by predicted complexity, and Probing-RAG [2] extends this idea with probes on the model’s internal states. Search-R1 [18] trains agentic retrieval with reinforcement learning, and RAG-Critic [4] adds an iterative critic. None of these methods conditions routing on hop distance in a typed graph or exposes a typed graph-lookup tool.

GraphRAG and multi-hop QA. Microsoft GraphRAG [6] and its successors [12, 13, 21] build entity–relation graphs at indexing time. Han et al. [15] and GraphRAG-Bench [31] report that GraphRAG does not outperform vector RAG on every query type. Our RQ2 results suggest one reason such verdicts differ: whether

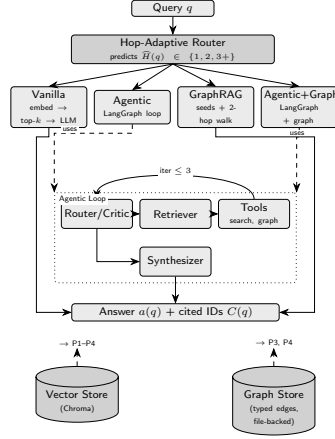


Fig. 1. The five pipelines and their shared components. Vanilla and GraphRAG make a single synthesis call. Agentic and Agentic+Graph run a router, retriever, tool, and critic loop for at most three iterations before synthesis. The adaptive pipeline (rule-based V1 or learned V2) selects one of the four base pipelines per query.

citations are scored on the retrieved set or on the answer. Of the multi-hop benchmarks MuSiQue [28], MultiHop-RAG [27], and 2WikiMultihopQA [16], we use the first and last to test RQ1 to RQ3 outside the requirements domain.

RAG for requirements traceability. LiSSA [10], TVR [23], and Graph-RAG for compliance checking [22] each evaluate a single regulated domain, without stratifying queries by hop depth and with a single judge.

Citation evaluation and judge reliability. ALCE [11] defined citation precision, recall, and F_1 for generated answers. Wallat et al. [29] show that ALCE-style correctness does not imply faithfulness, which motivates the use of LLM judges. RAGChecker [25] provides a single-judge framework, and self-preference bias [30] motivates judge ensembles. Feinstein and Cicchetti [9] describe the prevalence paradox of Cohen’s κ , and Gwet’s AC1 [14] is the correction we report alongside it. Our analysis for RQ4 adds a quantity this literature does not track, the agreement of a judge with its own earlier verdicts after the retrieval state or the date changes.

3 Method

3.1 Pipelines

All pipelines share the embedder, a ChromaDB vector store, a file-backed typed-edge graph, the GPT-5.4 generator, and a grounded synthesis prompt, so they differ only in how they retrieve (Fig. 1). (i) **Vanilla** retrieves the top 10 chunks

by dense similarity, passes the top 5 to the synthesizer, and makes one synthesis call. **(ii) Agentic** is a LangGraph [20] loop of router, retriever, and critic with a single `search_documents` tool and at most three iterations. **(iii) GraphRAG** takes 8 vector seeds, walks up to two hops in the typed graph (at most 30 nodes), and passes at most 15 chunks to one synthesis call. It is a local-walk retriever in the GraphRAG family rather than the community-summarisation system of Edge et al. [6]; we use “GraphRAG” as the family name. **(iv) Agentic+Graph** adds a typed `graph_lookup` tool to (ii). **(v) Adaptive** chooses among (i) to (iv) for each query. We evaluate a rule-based router (V1) and a learned router (V2), an out-of-fold logistic regression over 11 text features and 3 principal components of the query embedding that sends each predicted hop class to its empirically best pipeline.

3.2 Experimental Factors

Embedder. We use three models from three vendors at three dimensionalities: e5-small (384 dimensions), OpenAI text-embedding-3-small (1536), and BAAI bge-m3 (1024). The vector index is rebuilt per embedder; the graph is shared. **Traversal.** Besides the bounded undirected walk (at most two hops, at most 30 nodes), we run personalized PageRank from the same vector seeds. It is unbounded in radius and ranks nodes by random-walk mass, and all other settings are held fixed. **Corpus.** The main corpus is a synthetic DO-178C-style collection of 1,132 aerospace requirements in 30 modules (17 to 76 requirements per module; median length 59 tokens, 90th percentile 103). GPT-5.4 generated it from a hand-authored module taxonomy, and we release it with its link annotations. Retrieval traverses five traceability edge types (*derives_from*, *satisfies*, *references*, *verifies*, and *refines*); we exclude system-engineering edges such as allocation, which do not help retrieval. Two Wikipedia corpora test transfer: 200 MuSiQue [28] questions (67, 67, and 66 of two, three, and four hops, mapped to our three strata, with *REFERENCES* edges between consecutive supporting paragraphs) and 200 2WikiMultihopQA [16] questions, stratified by reasoning type: *comparison* questions are one-hop, *compositional* and *inference* questions are two-hop, and *bridge_comparison* questions are three-or-more-hop. **Judge.** Faithfulness is judged by GPT-5.4, GPT-4.1, and Gemini 3.7-flash (Section 4).

3.3 Queries and Gold Evidence

We build the DO-178C queries from the graph rather than by free-form prompting. We sample directed chains of one, two, and three to four requirement links over the five edge types and prompt GPT-5.4 to phrase one natural-language question that the chain answers. The nodes of the chain form the gold evidence, so the gold set grows with the stratum (on average 2.00, 3.00, and 4.68 IDs). This yields 296 queries (100, 99, and 97 per stratum), of which 85% cross module boundaries; 497 of the 655 chain edges are *references* links. The gold sets are structural rather than human-annotated: a sampled chain is a supporting path by construction,

but we do not verify that no other path answers the question. Citation recall is therefore exact and precision conservative, equally for all pipelines.

3.4 Metrics

We report citation precision, recall, and F_1 against the gold ID sets, following ALCE [11]. Cited IDs are parsed from the answer text with a parser anchored to the corpus vocabulary, which matches exact IDs and prefers the longest match. Hallucinated IDs that reuse a real module prefix count against precision, and incidental tokens such as SHA-256 or DO-254 are ignored. We report separately the context precision (the gold fraction of the chunks handed to the synthesizer) and the retrieval recall, since conflating context and citations changes which architecture appears to win (Section 5.2).

Each faithfulness judgment is a binary verdict, returned as strict JSON, on whether the answer is supported by the retrieved context. For agreement we report Cohen’s κ , Gwet’s AC1 [9, 14], raw agreement, and an exact McNemar test. For each faithfulness cell we report a Wilson 95% confidence interval and a Cochran–Armitage test for trend over hop depth. Three same-judge controls (test–retest, embedder swap, and re-judging after eleven weeks) use a paired subset of 300 tuples (Section 5.4).

4 Experimental Setup

Runs. On DO-178C, we run 5 pipelines $\times$ 296 queries $\times$ 3 seeds, giving 4,440 runs under each of two embedders (e5-small, which we call v_2 , and text-embedding-3-small, v_3). We add 1,480 runs at one seed under bge-m3 (v_4) and a 296-run personalized PageRank arm under the v_3 embedder. On MuSiQue, we run vanilla, GraphRAG, and agentic+graph on 200 queries at one seed (600 runs, v_3 embedder), since RQ2 needs only these three, and extend the GraphRAG arm to three seeds to match DO-178C’s per-stratum sample. On the same queries we add 400 runs for two mitigations and a 200-run GraphRAG control with distractor edges (Section 5.1). On 2WikiMultihopQA, we run vanilla and GraphRAG on 200 queries at three seeds (1,200 runs, v_3 embedder).

Faithfulness judging. On DO-178C, GPT-5.4 and GPT-4.1 judge two disjoint stratified batches of 300 rows each (60 per pipeline; seeds 42 and 43), the second eleven weeks after the first. Each batch is fixed to the same 300 (query, pipeline, repeat) tuples across embedders, so embedder differences are computed on identical items, and batches judged on different dates are never pooled. In a later pass, GPT-4.1 and Gemini 3.7-flash judge every row of the DO-178C matrices and of the Wikipedia corpora. Gemini judged four days after GPT-4.1 on DO-178C and one day after it on the Wikipedia corpora, in both cases on the same stored answer and context strings, so the pairing is exact at the row level. Two same-input controls re-judge 300 v_2 tuples with GPT-5.4, once on the same day (test–retest) and once eleven weeks later. A generator-swap control

Table 1. Citation F_1 by hop stratum in four settings. v2 and v3: DO-178C with the e5-small and text-embedding-3-small embedders (4,440 runs each). MuSiQue: 200 queries with the v3 embedder (600 runs; vanilla, agentic+graph, and GraphRAG only). 2Wiki: 200 2WikiMultihopQA queries at three seeds (1,200 runs; vanilla and GraphRAG only). Bold marks the best pipeline per setting and stratum.

System	v2 main (local)			v3 main (Azure)			MuSiQue (Azure)			2Wiki (Azure)		
	1h	2h	3+h	1h	2h	3+h	1h	2h	3+h	1h	2h	3+h
vanilla	.757	.712	.089	.756	.689	.195	.742	.415	.277	.990	.778	.678
agentic	.541	.414	.175	.533	.430	.203	—	—	—	—	—	—
agentic-graph	.566	.415	.219	.547	.421	.251	.457	.296	.249	—	—	—
graphrag	.751	.720	.172	.745	.707	.253	.841	.632	.463	.969	.946	.938
adaptive	.630	.499	.181	.651	.492	.254	—	—	—	—	—	—

re-synthesizes 332 v3 rows with GPT-4.1 on frozen retrievals and judges them with both judges.

Statistical protocol. We compute 95% BCa bootstrap intervals [5] with 1,000 resamples paired at the query level, Wilcoxon signed-rank tests with Holm correction over the 30 contrasts of pipeline pair and stratum [3, 19], and Cliff’s δ , which we treat as negligible below 0.147. We call two pipelines significantly different only when the BCa interval excludes zero, the Holm-adjusted p is below 0.05, and $|\delta| \geq 0.147$. Tests pool the three seeds; we report the spread across seeds separately (Section 5.1).

Reproducibility. The original graph store was populated before this study and contains edges beyond those annotated in the released corpus. Loading the release with our ETL recovers 91.0% of the edges and 82.8% of the chains behind our queries. Every gold set is nevertheless exact, since the query manifest is released with its full chains. The v2 and v3 graph arms walked the original store, while the bge-m3 matrix and the PageRank arm walked a file-backed store built from the released annotations, which is also what a reproduction uses. To check that the change of graph does not drive our comparisons, we rebuilt every v3 GraphRAG context on the file-backed store without generation. The vector seeds are identical for 295 of the 296 queries, the two contexts have a Jaccard overlap of 0.89, and context precision (0.131 against 0.129) and retrieval recall (0.720 against 0.724) differ by at most 0.01 in every stratum. We did not regenerate answers on the new store, so the citation side of this check is untested. We use the Azure snapshots `gpt-5.4-2026-03-05` (generator and judge) and `gpt-4.1-2025-04-14` (judge) in reasoning mode, with a completion cap of 8,192 tokens. The MuSiQue subgraph (3,996 paragraph chunks and 399 *REFERENCES* edges) is built deterministically from the `dgslibisey` mirror (validation split, seed 20260511), and the 2WikiMultihopQA subgraph (2,000 chunks and 332 edges) from the `voidful` mirror (seed 20260816) by the same procedure.

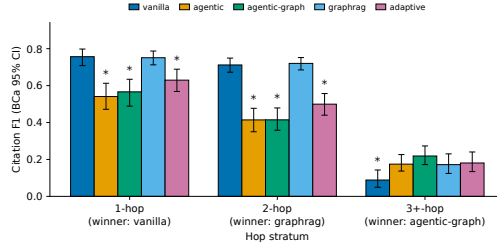


Fig. 2. Citation F_1 by hop stratum on the v2 matrix (DO-178C, e5-small), with 95% BCa intervals. An asterisk marks a pipeline that differs significantly from the best pipeline in its stratum (Holm-adjusted Wilcoxon $p < 0.05$ and $|\delta| \geq 0.147$).

5 Results

5.1 Citation Quality by Hop Depth (RQ1)

Table 1 reports citation F_1 by stratum in four settings. On DO-178C, vanilla and GraphRAG are statistically tied on one- and two-hop queries (Wilcoxon $p = 0.12$ and 0.65 under the v2 embedder, with negligible δ), and the agentic loop costs 0.19 to 0.30 F_1 on these strata. On queries of three or more hops the ordering reverses. Vanilla falls to last place, and the pipelines that use the graph lead; under the v2 embedder agentic+graph reaches 0.219 against 0.089 for vanilla (Holm $p < 10^{-6}$). Under the v3 embedder the ordering is the same, but no pair at this depth passes the joint criterion. Under bge-m3 the pattern recurs (Table 2): vanilla is last at three or more hops under all three embedders, and under bge-m3 both graph pipelines outperform it by 0.100 and 0.091 (Holm $p = 2.1 \times 10^{-4}$ and 1.4×10^{-3}), while the one- and two-hop contrasts remain negligible.

On MuSiQue, GraphRAG is best in every stratum (F_1 of 0.841 , 0.632 , and 0.463), with $p \leq 5.1 \times 10^{-4}$ and δ between 0.27 and 0.43 against vanilla. Because the MuSiQue *REFERENCES* edges connect only gold paragraphs, we also ran GraphRAG on a graph with additional edges between consecutive distractor paragraphs. GraphRAG then loses 0.02 , 0.05 , and 0.09 F_1 per stratum and still outperforms vanilla in every stratum ($p \leq 0.014$), so its advantage on MuSiQue does not depend on gold-only edges. On 2WikiMultihopQA, the two pipelines are tied on one-hop queries ($p = 0.15$), and GraphRAG leads on two-hop and deeper queries by 0.168 and 0.260 ($\delta = 0.48$ and 0.87). This is the DO-178C crossover again, on a graph we did not build.

Seed variation is small by comparison. The mean F_1 of each pipeline changes by at most 0.031 between seeds, every significant ordering holds in each seed separately, and the only changes in rank occur between vanilla and GraphRAG on one- and two-hop queries, which we already report as ties. Regarding RQ1, the best pipeline depends on hop depth and on the corpus, but not on the embedder or the seed.

Table 2. Citation F_1 by hop stratum on DO-178C under three embedders from three vendors (e5-small, 384 dimensions; text-embedding-3-small, 1536; bge-m3, 1024). Bold marks the best pipeline per embedder and stratum.

System	e5-small (384d)			3-small (1536d)			bge-m3 (1024d)		
	1h	2h	3+h	1h	2h	3+h	1h	2h	3+h
vanilla	.757	.712	.089	.756	.689	.195	.769	.691	.114
agentic	.541	.414	.175	.533	.430	.203	.548	.404	.156
agentic-graph	.566	.415	.219	.547	.421	.251	.546	.431	.206
graphrag	.751	.720	.172	.745	.707	.253	.742	.728	.214
adaptive	.630	.499	.181	.651	.492	.254	.672	.478	.202

Table 3. Context and citation precision of GraphRAG in six settings. Slots: chunks passed to the synthesizer. IDs: IDs cited in the answer. Gain: citation precision divided by context precision.

Setting	<i>retrieved context</i>		<i>answer citations</i>		
	slots	prec.	IDs	prec.	gain
DO-178C, e5-small	14.9	0.120	4.9	0.480	4.0×
DO-178C, 3-small	14.9	0.129	5.0	0.493	3.8×
DO-178C, bge-m3	14.8	0.121	4.7	0.490	4.1×
DO-178C, PPR walk	14.9	0.131	4.9	0.509	3.9×
MuSiQue	11.2	0.227	3.4	0.654	2.9×
2WikiMultihopQA	11.4	0.230	2.5	0.975	4.2×

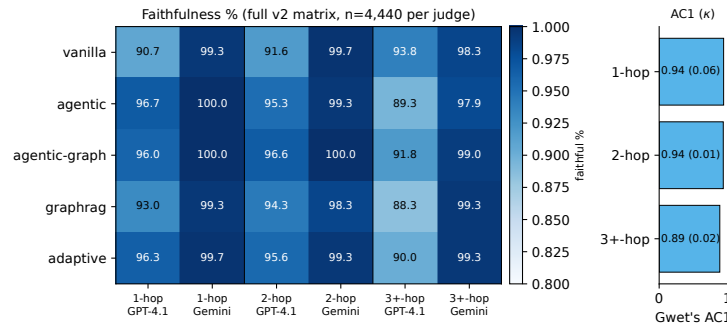


Fig. 3. Faithfulness on the full v2 matrix (DO-178C, e5-small; 4,440 rows, about 300 per cell), judged by GPT-4.1 and Gemini 3.7-flash on the same rows. Left: percentage of answers judged faithful. Right: Gwet's AC1 between the two judges per stratum, with Cohen's κ in parentheses. The colour scale starts at 80%.

5.2 Context Versus Answer Citations (RQ2)

Table 3 reports the same measurement in six settings, covering three embedders, three corpora, and two traversals. GraphRAG's walk fills 11 to 15 context slots at

a context precision of 0.12 to 0.23, so four or five of every six chunks it retrieves are not gold. The synthesizer then cites only 2.5 to 5.0 IDs per answer, at a citation precision of 0.48 to 0.98. In every setting, the precision of the citations is 2.9 to 4.2 times that of the context. The F_1 advantage of GraphRAG is therefore a recall effect: the walk retrieves gold chunks that dense retrieval misses, and the synthesizer does not cite most of the irrelevant chunks that come with them.

An evaluation that scores the retrieved set as the attribution set, as some GraphRAG comparisons do, sees only the first step and reaches the opposite ranking. By context precision, GraphRAG is below the vanilla baseline in every setting (0.12 to 0.23, against 0.27 to 0.37 for vanilla). By citation F_1 , it is the best pipeline in every setting, with scores of 0.551, 0.571, 0.565, 0.585, 0.646, and 0.951 in the row order of Table 3.

Two controls test whether this depends on our design. The first replaces the two-hop walk with personalized PageRank over the same graph. Context precision is 0.131 with PageRank and 0.129 with the original walk (0.131 for the walk rebuilt on the same file-backed graph; Section 4). The precision gain from context to citations is 3.9 and 3.8 times, and the difference in citation F_1 is 0.016 with $\delta = 0.036$, which our criterion treats as negligible. The second control applies two mitigations on MuSiQue. Reranking the walked neighbours by similarity to the query and keeping the top five changes little (context precision from 0.227 to 0.232, citation $F_1 + 0.010$, $p = 0.34$). Reducing the traversal budget from 30 to 10 nodes and the context from 15 to 8 chunks lowers citation F_1 by 0.126 ($p = 2.6 \times 10^{-11}$) and also lowers context precision, to 0.194. Both outcomes fit a synthesizer that already filters: an upstream filter adds little, and a smaller budget removes the recall the advantage depends on. Regarding RQ2, the comparison reverses with the measurement point in all six settings.

5.3 Faithfulness and Hop Depth (RQ3)

On 300-row judged subsets, faithfulness declines with hop depth on DO-178C but not on the Wikipedia chains. This difference disappears at adequate power. The MuSiQue arm had 67 rows per stratum. Its drop from one-hop to three-or-more-hop queries (9.2 percentage points) was larger than the 4.7-point drop that is significant on DO-178C, but detecting it with 80% power requires about 210 rows per cell. With three seeds ($n = 600$), the decline is significant under both judges, from 0.930 to 0.876 to 0.823 under GPT-4.1 ($p = 1.1 \times 10^{-3}$) and from 0.955 to 0.945 to 0.869 under Gemini ($p = 1.2 \times 10^{-3}$). On 2WikiMultihopQA, judged in full ($n = 1,200$), faithfulness also declines ($p = 5.0 \times 10^{-2}$ and 4.8×10^{-6}). The decline therefore replicates on all three corpora under two vendors.

What distinguishes the pipelines is whether they expand the context. Vanilla retrieves a fixed top five and adds nothing, and under GPT-4.1 it shows no significant drop from one-hop to three-or-more-hop queries. Its change is -1.5 points on DO-178C ($n = 2,072$; $p = 0.16$, 0.99, and 0.76 for the three embedders) and $+1.1$ points on 2WikiMultihopQA ($p = 0.66$). The four expanding pipelines drop by 6.3 and 6.7 points on the same corpora (Fig. 3). A logistic regression of faithfulness on hop depth, an expansion indicator, and their interaction turns

Table 4. Judge agreement. Top: the same judge re-judging identical answer and context strings, so only the date differs. Middle: the same judge after an embedder swap changes the input. Bottom: GPT-4.1 against Gemini 3.7-flash on identical rows; the last column gives the fraction judged faithful by each.

<i>Same judge, frozen input (answer + context identical)</i>				
GPT-5.4, re-judged same day				$\kappa = 0.764$, raw agr. 0.88
GPT-5.4, re-judged 11 wk later				$\kappa = 0.138$, raw agr. 0.56
GPT-4.1, re-judged 11 wk later	$\kappa = -0.05$	-0.00	$(v2/v3)$	raw agr. 0.48 / 0.51
GPT-4.1, full matrix vs. May				$\kappa = -0.046$, raw agr. 0.48, +43 pp
<i>Same judge, input changed by embedder swap</i>				
GPT-5.4 (v2 vs v3)				$\kappa = 0.137$, raw agr. 0.59
GPT-4.1 (v2 vs v3)				$\kappa = 0.480$, raw agr. 0.74
<i>Cross-vendor, same rows (Gemini 3.7-flash vs. GPT-4.1)</i>				
<i>Setting</i>	raw	κ	AC1	faithful (G / 4.1)
DO-178C e5-small ($n=4,440$)	0.929	0.029	0.923	0.993 / 0.933
DO-178C 3-small ($n=4,440$)	0.913	0.125	0.903	0.980 / 0.917
DO-178C bge-m3 ($n=1,480$)	0.939	0.047	0.935	0.991 / 0.945
DO-178C PPR walk ($n=296$)	0.926	0.058	0.919	0.983 / 0.936
MuSiQue, 3 seeds ($n=600$)	0.850	0.172	0.817	0.923 / 0.877
MuSiQue rerank ($n=200$)	0.820	0.171	0.772	0.935 / 0.825
MuSiQue budget cap ($n=200$)	0.875	0.419	0.841	0.875 / 0.880
2WikiMultihopQA ($n=1,200$)	0.899	0.435	0.877	0.888 / 0.914

this contrast into a single test. The interaction is negative and significant on the pooled DO-178C matrix ($b = -0.599$, $p = 1.0 \times 10^{-7}$, $n = 10,360$), and negative but not significant on 2WikiMultihopQA ($b = -0.286$, $p = 0.27$), whose arm is one twentieth the size.

Under Gemini the ordering is the same, but the test does not resolve. Near the ceiling, vanilla itself shows a slope ($b = -0.729$, $p = 0.012$), and the interaction cannot be separated from it ($p = 0.11$). The expanding pipelines still drop more than vanilla (2.8 against 1.5 points on DO-178C, 12.7 against 7.7 on 2WikiMultihopQA). A lenient judge preserves the ranking but not the power to test it (Section 5.4). Not every expanding pipeline declines in every cell: GraphRAG is marginal under bge-m3 ($p = 0.065$), and the agentic pipelines decline in four of six embedder and pipeline combinations. We have not re-run the GPT-5.4 judge at this sample size. Regarding RQ3, faithfulness declines with hop depth on every corpus, and the decline is steeper for pipelines that expand their context.

5.4 Judge Stability (RQ4)

Table 4 separates instability on fixed inputs from instability after the retrieval state changes. *Fixed inputs.* Re-judging the same 300 stored answer and context

pairs gives GPT-5.4 a self-agreement of $\kappa = 0.76$ (raw agreement 0.88) on the same day, but only $\kappa = 0.14$ (raw agreement 0.56) eleven weeks later, with 15 points more answers judged faithful. This rejects a stationary judge (binomial $p < 10^{-44}$). GPT-4.1 on the same tuples agrees with its own earlier verdicts only at chance level ($\kappa = -0.05$ and -0.00 for v2 and v3) and judges 41 points more answers faithful. With fixed inputs, this is drift of the judge alone. *Changed retrieval state.* After an embedder swap, which changes the context and the answer for 94% of the tuples, GPT-5.4 changes 41% of its verdicts ($\kappa = 0.137$) and GPT-4.1 changes 26% ($\kappa = 0.48$). The judge that is more stable under the embedder swap is thus the less stable over time. *Across vendors.* On the same rows, Gemini 3.7-flash is much more lenient than GPT-4.1 on all four DO-178C settings (0.98 to 0.99 against 0.92 to 0.95 faithful; McNemar $p < 10^{-9}$). They are indistinguishable on the budget-capped MuSiQue arm (0.875 against 0.880, $p = 1.00$), and Gemini is stricter on 2WikiMultihopQA (0.888 against 0.914, $p = 6.2 \times 10^{-3}$). Strictness is therefore a property of the judge and corpus together, and a calibration offset measured on one corpus cannot be reused on another. The same eight settings show the prevalence paradox [9] within one pair of judges. Raw agreement ranges only from 0.82 to 0.94, and Gwet’s AC1 follows it (0.77 to 0.94), whereas κ ranges from 0.03 to 0.44, ordered almost exactly by the distance of faithfulness from the ceiling. On 2WikiMultihopQA the two judges agree less often than on DO-178C (0.899 against 0.93), yet their κ is fifteen times higher (0.435 against 0.029). Leniency and trend are nevertheless separate properties. Although Gemini is near the ceiling on DO-178C, it reproduces the decline with hop depth on both DO-178C matrices ($p = 4.3 \times 10^{-3}$ and $p < 10^{-16}$) and on both Wikipedia corpora. The ordering a judge induces replicates across vendors, while its strictness does not. *Between judges.* Inter-judge agreement halves under the embedder swap (κ from 0.30 to 0.17; McNemar $p < 0.001$), and on the 300-row MuSiQue batch GPT-5.4 and GPT-4.1 agreed only at chance (raw agreement 0.48, AC1 0.08), a real disagreement rather than a prevalence effect. We therefore re-judged the full matrix within one judging period (Section 5.3), and no comparison we report spans judging dates. Regarding RQ4, judges are not stable over time, across retrieval states, or across vendors, although the hop-wise ordering they induce is.

6 Discussion

Why expansion lowers faithfulness. The synthesizer leaves most irrelevant chunks out of its citations (Section 5.2), but they can still influence the answer. Because faithfulness is judged against the full retrieved context, every chunk the walk adds is one more chunk the answer can be judged unfaithful to. This mechanism does not depend on what the edges connect, consistent with the decline on both typed DO-178C links and Wikipedia paragraph links. The cost appears to scale with the number of added chunks rather than with their relevance, since reranking them by relevance changes nothing (Section 5.2).

Implications for LLM judges. Since judges change with the retrieval state and the date (Section 5.4), single-judge comparisons of retrieval modules are conditional on both, and ensembles help only partially. At a minimum, faithfulness should be compared on paired tuples under a fixed embedder and judged within one period.

Threats to validity. *Construct validity.* Citation F_1 does not measure the quality of the reasoning, so we pair it with faithfulness judgments, which are themselves subject to bias [29,30]. We quantify this bias directly (Section 5.4) and use a third judge from a different vendor on every matrix. *External validity.* The DO-178C corpus is a single synthetic dataset, generated by the same model family that answers and judges, and its absolute F_1 levels may not transfer to other regulated domains. When GPT-4.1 re-synthesizes 332 answers on frozen retrievals, the pipeline ordering and the faithfulness decline both hold, so neither result is specific to GPT-5.4. The corpus, however, is still written by that family. The two Wikipedia corpora, judged in full by two vendors, are our main check of external validity; both reproduce the RQ1 crossover, although only DO-178C includes all five pipelines. Our GraphRAG pipeline is one local-walk implementation. Replacing the walk with personalized PageRank leaves the RQ2 result unchanged, but a full third-party GraphRAG system may flood or filter its context differently. The two unsupervised mitigations we tried did not help, and a learned evidence selector conditioned on the question remains to be tested. *Statistical conclusion validity.* Per-stratum samples for the citation metrics (97 to 100 queries on DO-178C, 66 to 67 on Wikipedia) are at the lower end of BCa stability under strong skew, but percentile intervals give the same conclusions. Faithfulness cells hold about 300 rows after full-matrix judging. The hop-by-expansion interaction is significant on DO-178C, but on the much smaller 2WikiMultihopQA arm we can report only its direction.

Artifact availability. The corpus with its link annotations, the query manifest with full chains, the code, and every scored and judged CSV file are released under CC-BY-4.0 (links withheld for review). The Wikipedia subsets are rebuilt by seeded scripts.

7 Conclusion

In this paper, we compared five RAG pipelines for multi-hop requirements traceability while varying the embedder, corpus, graph traversal, and judge. The citation measurement point reverses the GraphRAG comparison in all six settings, the best pipeline per hop depth is stable across embedders and seeds, and the faithfulness decline follows context expansion on all three corpora. LLM judges drift on identical inputs but agree on the hop-wise ordering. RAG comparisons should therefore state their measurement point and report results across these factors; testing a learned evidence selector and a third-party GraphRAG system are natural next steps.

References

1. Asai, A., Wu, Z., Wang, Y., Sil, A., Hajishirzi, H.: Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In: Proc. International Conference on Learning Representations (ICLR) (2024)
2. Baek, I., Chang, H., Kim, B., Lee, J., Lee, H.: Probing-RAG: Self-probing to guide language models in selective document retrieval. In: Findings of NAACL (2025), arXiv:2410.13339
3. Berg-Kirkpatrick, T., Burkett, D., Klein, D.: An empirical investigation of statistical significance in NLP. In: Proc. EMNLP-CoNLL (2012)
4. Dong, G., Jin, J., Li, X., Zhu, Y., Dou, Z., Wen, J.R.: RAG-Critic: Leveraging automated critic-guided agentic workflow for retrieval augmented generation. In: Proc. 63rd Annual Meeting of the Association for Computational Linguistics (ACL): Long Papers. pp. 3551–3578 (2025), anthology ID: 2025.acl-long.179
5. Du, W.: When +1% is not enough: A paired bootstrap protocol for evaluating small improvements. arXiv preprint arXiv:2511.19794 (2025)
6. Edge, D., Trinh, H., Cheng, N., Bradley, J., Chao, A., Mody, A., Truitt, S., Larson, J.: From local to global: A graph RAG approach to query-focused summarization. arXiv preprint arXiv:2404.16130 (2024)
7. European Union Aviation Safety Agency (EASA): Notice of proposed amendment NPA 2025-07: Detailed specifications on AI trustworthiness. EASA (2025)
8. Federal Aviation Administration: AI safety roadmap, version 1. FAA (2024)
9. Feinstein, A.R., Cicchetti, D.V.: High agreement but low kappa: I. the problems of two paradoxes. *Journal of Clinical Epidemiology* **43**(6), 543–549 (1990)
10. Fuchs, D., Hey, T., Keim, J., Liu, H., Ewald, N., Thirolf, T., Koziol, A.: LiSSA: Toward generic traceability link recovery through retrieval-augmented generation. In: Proc. IEEE/ACM 47th International Conference on Software Engineering (ICSE) (2025)
11. Gao, T., Yen, H., Yu, J., Chen, D.: Enabling large language models to generate text with citations. In: Proc. EMNLP (2023), arXiv:2305.14627
12. Guo, Z., Xia, L., Yu, Y., Ao, T., Huang, C.: LightRAG: Simple and fast retrieval-augmented generation. In: Findings of the Association for Computational Linguistics: EMNLP 2025. pp. 10746–10761 (2025), arXiv:2410.05779
13. Gutiérrez, B.J., Shu, Y., Qi, W., Zhou, S., Su, Y.: HippoRAG 2: From RAG to memory: Non-parametric continual learning for large language models. arXiv preprint arXiv:2502.14802 (2025)
14. Gwet, K.L.: Computing inter-rater reliability and its variance in the presence of high agreement. *British Journal of Mathematical and Statistical Psychology* **61**(1), 29–48 (2008)
15. Han, H., Ma, L., Wang, Y., Shomer, H., Lei, Y., Qi, Z., Guo, K., Hua, Z., Long, B., Liu, H., Aggarwal, C.C., Tang, J.: RAG vs. GraphRAG: A systematic evaluation and key insights. arXiv preprint arXiv:2502.11371 (2025)
16. Ho, X., Duong Nguyen, A.K., Sugawara, S., Aizawa, A.: Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps. In: Proc. 28th International Conference on Computational Linguistics (COLING). pp. 6609–6625 (2020)
17. Jeong, S., Baek, J., Cho, S., Hwang, S.J., Park, J.C.: Adaptive-RAG: Learning to adapt retrieval-augmented large language models through question complexity. In: Proc. NAACL (2024)

18. Jin, B., Zeng, H., Yue, Z., Yoon, J., Arik, S., Wang, D., Zamani, H., Han, J.: Search-R1: Training LLMs to reason and leverage search engines with reinforcement learning. In: Proc. 2nd Conference on Language Modeling (COLM) (2025), arXiv:2503.09516
19. Koehn, P.: Statistical significance tests for machine translation evaluation. In: Proc. EMNLP (2004)
20. LangChain, Inc.: LangGraph: A library for building stateful, multi-actor applications with LLMs. Software, <https://github.com/langchain-ai/langgraph> (2024)
21. Luo, H., E, H., Chen, G., Lin, Q., Guo, Y., Xu, F., Kuang, Z., Song, M., Wu, X., Zhu, Y., Tuan, L.A.: Graph-R1: Towards agentic GraphRAG framework via end-to-end reinforcement learning. In: Proc. International Conference on Machine Learning (ICML) (2026), arXiv:2507.21892
22. Masoudifard, A., Mowlavi Sorond, M., Madadi, M., Sabokrou, M., Habibi, E.: Leveraging graph-RAG and prompt engineering to enhance LLM-based automated requirement traceability and compliance checks. arXiv preprint arXiv:2412.08593 (2024)
23. Niu, F., Pan, R., Briand, L.C., Hu, H., Koravadi, K.: TVR: Automotive system requirement traceability validation and recovery through retrieval-augmented generation. arXiv preprint arXiv:2504.15427 (2025)
24. RTCA, Inc.: DO-178C: Software considerations in airborne systems and equipment certification. Tech. rep., RTCA (2011)
25. Ru, D., Qiu, L., Hu, X., Zhang, T., Shi, P., et al.: RAGChecker: A fine-grained framework for diagnosing retrieval-augmented generation. In: Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track (2024), arXiv:2408.08067
26. SAE International: ARP4754A: Guidelines for development of civil aircraft and systems. Tech. rep., SAE International (2010)
27. Tang, Y., Yang, Y.: MultiHop-RAG: Benchmarking retrieval-augmented generation for multi-hop queries. In: Proc. 1st Conference on Language Modeling (COLM) (2024), arXiv:2401.15391
28. Trivedi, H., Balasubramanian, N., Khot, T., Sabharwal, A.: MuSiQue: Multihop questions via single-hop question composition. Transactions of the Association for Computational Linguistics (TACL) **10**, 539–554 (2022)
29. Wallat, J., Heuss, M., de Rijke, M., Anand, A.: Correctness is not faithfulness in RAG attributions. arXiv preprint arXiv:2412.18004 (2024)
30. Wataoka, K., Takahashi, T., Ri, R.: Self-preference bias in LLM-as-a-judge. In: NeurIPS Safe Generative AI Workshop (2024), arXiv:2410.21819
31. Xiang, Z., Wu, C., Zhang, Q., Chen, S., Hong, Z., Huang, X., Su, J.: When to use graphs in RAG: A comprehensive analysis for graph retrieval-augmented generation. In: Proc. International Conference on Learning Representations (ICLR) (2026), arXiv:2506.05690
32. Yan, S.Q., Gu, J.C., Zhu, Y., Ling, Z.H.: Corrective retrieval augmented generation. arXiv preprint arXiv:2401.15884 (2024)